Bayesian Optimization Report: Function 3

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.635	Medium
2	0.636	Medium
3	0.637	Medium
4	0.639	Medium
5	0.666	Medium
6	0.755	Good
7	0.641	Medium
8	0.748	Good
9	0.778	Good
10	0.777	Good
11	0.775	Good
12	0.774	Good
13	0.772	Good
14	0.771	Good
15	0.769	Good
16	0.768	Good
17	0.767	Good
18	0.765	Good
19	0.763	Good
20	0.763	Good
21	0.761	Good
22	0.760	Good
23	0.759	Good
24	0.758	Good
25	0.757	Good
26	0.755	Good
27	0.754	Good
28	0.753	Good
29	0.752	Good
30	0.751	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	1.0000	1.0000	1.0000
2	0.9798	0.9798	0.9798
3	0.9608	0.9608	0.9608
4	0.9428	0.9428	0.9428
5	0.9258	0.9258	0.9258
6	0.9097	0.9097	0.9097
7	0.8944	0.8944	0.8944
8	0.8799	0.8799	0.8799
9	0.6906	0.6805	0.7346
10	0.6802	0.6708	0.7128
11	0.6696	0.6611	0.7018
12	0.6606	0.6516	0.6979
13	0.6518	0.6428	0.6894
14	0.6429	0.6339	0.6873
15	0.6344	0.6256	0.6627
16	0.6269	0.6173	0.6601
17	0.6201	0.6100	0.6646
18	0.6132	0.6028	0.6486
19	0.6059	0.5963	0.6397
20	0.5984	0.5891	0.6544
21	0.5931	0.5832	0.6264
22	0.5873	0.5770	0.6142
23	0.5814	0.5709	0.6165
24	0.5752	0.5654	0.6036
25	0.5696	0.5594	0.6163
26	0.5648	0.5536	0.5930
27	0.5599	0.5487	0.6010
28	0.5546	0.5432	0.5827
29	0.5497	0.5385	0.5709
30	0.5462	0.5332	0.5896

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.42197268 0.75044962 0.36329243]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.635	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
2	[0.3719064 0.85381306 0.43489009]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.636	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
3	[0.48877008 0.76383849 0.4262724]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.637	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
4	[0.35210286 0.84927376 0.41640945]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.639	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
5	[0.39470462 0.70453651 0.37632747]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.666	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
6	[0.38020921 0.69477564 0.38168145]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.755	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
7	[0.345238 0.74754116 0.43479719]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.641	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
8	[0.47746466 0.72531824 0.40664492]	-0.0902367444591603	Matern(ν=0.5)	UCB	0.748	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
9	[0.5833868 0.86050746 0.43076552]	-0.0692695416428224	Matern(ν=0.5)	UCB	0.778	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
10	[0.4675839 0.5947012 0.46024753]	-0.06421874824605611	Matern(ν=0.5)	UCB	0.777	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}

11	[0.53158704 0.65460408 0.36857069]	-0.061821137409202365	Matern(ν=0.5)	UCB	0.775	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
12	[0.57278299 0.65081138 0.34883682]	-0.060811037047083794	Matern(ν=0.5)	UCB	0.774	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
13	[0.52657656 0.69723764 0.43044713]	-0.06368475745085833	Matern(ν=0.5)	UCB	0.772	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
14	[0.48293264 0.61264156 0.38189075]	-0.06275126728331082	Matern(ν=0.5)	UCB	0.771	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
15	[0.56548254 0.72676817 0.37420625]	-0.06252641482498622	Matern(ν=0.5)	UCB	0.769	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
16	[0.58577173 0.79166405 0.36866093]	-0.06437937057 13771	Matern(ν=0.5)	UCB	0.768	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
17	[0.43410287 0.55451491 0.41518091]	-0.06380983848082811	Matern(ν=0.5)	UCB	0.767	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
18	[0.52608631 0.62664276 0.46268163]	-0.06239579470190049	Matern(ν=0.5)	UCB	0.765	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
19	[0.50536119 0.65813106 0.40348305]	-0.06301446443251191	Matern(ν=0.5)	UCB	0.763	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
20	[0.66146229 0.74804094 0.43430987]	-0.06065673766568236	Matern(ν=0.5)	UCB	0.763	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}

21	[0.50764513 0.56588798 0.42089724]	-0.06151898231107911	Matern(ν=0.5)	UCB	0.761	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
22	[0.56143812 0.6562203 0.42845021]	-0.061035898995114746	Matern(ν=0.5)	UCB	0.760	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
23	[0.53286931 0.62052035 0.38593055]	-0.061146142918776736	Matern(ν=0.5)	UCB	0.759	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
24	[0.54305991 0.69954856 0.36217524]	-0.06260027543440816	Matern(ν=0.5)	UCB	0.758	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
25	[0.53476161 0.6841977 0.3592447]	-0.06251539653490235	Matern(ν=0.5)	UCB	0.757	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
26	[0.5421195 0.70447646 0.3809647]	-0.0627552173389252	Matern(ν=0.5)	UCB	0.755	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
27	[0.53149468 0.6668693 0.44842017]	-0.06264344387438148	Matern(ν=0.5)	UCB	0.754	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
28	[0.54196886 0.69383384 0.40175962]	-0.062445035519918646	Matern(ν=0.5)	UCB	0.753	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
29	[0.54917462 0.67871257 0.39921187]	-0.06167768239552873	Matern(ν=0.5)	UCB	0.752	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}
30	[0.55551406 0.72444167 0.42792884]	-0.06313137455831064	Matern(ν=0.5)	UCB	0.751	{'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}}

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	1.968707
EI	0.009924
PI	0.024690
THOMPSON	1.638288
KG	0.009924
ENTROPY	7.192371
PORTFOLIO	3.065032

Iteration 2

Method	Max Value
UCB	0.750385
EI	0.008895
PI	0.022850
THOMPSON	1.863931
KG	0.008895
ENTROPY	7.271078
PORTFOLIO	2.681894

Iteration 3

Method	Max Value
UCB	0.737124
EI	0.007983
PI	0.021154
THOMPSON	2.082412
KG	0.007983
ENTROPY	6.855308
PORTFOLIO	2.574498

Iteration 4

Method	Max Value
UCB	0.724821
EI	0.007178
PI	0.019597

THOMPSON	1.828641
KG	0.007178
ENTROPY	6.582869
PORTFOLIO	2.414316

Iteration 5

Method	Max Value
UCB	0.741829
EI	0.006722
PI	0.018891
THOMPSON	1.903602
KG	0.006722
ENTROPY	7.260965
PORTFOLIO	2.305275

Iteration 6

Method	Max Value
UCB	0.826822
EI	0.006859
PI	0.019826
THOMPSON	1.633991
KG	0.006859
ENTROPY	6.647583
PORTFOLIO	2.394870

Iteration 7

Method	Max Value
UCB	0.690294
EI	0.005246
PI	0.015582
THOMPSON	1.599298
KG	0.005246
ENTROPY	6.041220
PORTFOLIO	1.964993

Iteration 8

Method	Max Value
UCB	0.792139
EI	0.005519
PI	0.016829
THOMPSON	2.810843
KG	0.005519
ENTROPY	7.880061
PORTFOLIO	2.116658

Iteration 9

Method	Max Value
UCB	1.096700
EI	0.009705
PI	0.033911
THOMPSON	1.935467
KG	0.009705
ENTROPY	6.245027
PORTFOLIO	1.822571

Iteration 10

Method	Max Value
UCB	1.056185
EI	0.008184
PI	0.029162
THOMPSON	2.285378
KG	0.008184
ENTROPY	6.262635
PORTFOLIO	1.856717

Iteration 11

Method	Max Value
UCB	1.066103
EI	0.007751
PI	0.028915

THOMPSON	1.401297
KG	0.007751
ENTROPY	5.532842
PORTFOLIO	1.624867

Iteration 12

Method	Max Value
UCB	1.075881
EI	0.008014
PI	0.029996
THOMPSON	1.619871
KG	0.008014
ENTROPY	5.528784
PORTFOLIO	1.522051

Iteration 13

Method	Max Value
UCB	1.013832
EI	0.005547
PI	0.022169
THOMPSON	1.670165
KG	0.005547
ENTROPY	4.574626
PORTFOLIO	1.489604

Iteration 14

Method	Max Value
UCB	1.016648
EI	0.005502
PI	0.022267
THOMPSON	1.641969
KG	0.005502
ENTROPY	4.911306
PORTFOLIO	1.361571

Iteration 15

Method	Max Value
UCB	1.011705
EI	0.005223
PI	0.021489
THOMPSON	1.892054
KG	0.005223
ENTROPY	5.764613
PORTFOLIO	1.350957

Iteration 16

Method	Max Value
UCB	1.003161
EI	0.004804
PI	0.020236
THOMPSON	1.611111
KG	0.004804
ENTROPY	4.311254
PORTFOLIO	1.416493

Iteration 17

Method	Max Value
UCB	0.998835
EI	0.004515
PI	0.019419
THOMPSON	2.129654
KG	0.004515
ENTROPY	4.374142
PORTFOLIO	1.287598

Iteration 18

Method	Max Value
UCB	0.994585
EI	0.004535
PI	0.019373

THOMPSON	1.574402
KG	0.004535
ENTROPY	4.125285
PORTFOLIO	1.126805

Iteration 19

Method	Max Value
UCB	0.961623
EI	0.003430
PI	0.015584
THOMPSON	1.484463
KG	0.003430
ENTROPY	4.909009
PORTFOLIO	1.198644

Iteration 20

Method	Max Value
UCB	1.019619
EI	0.005261
PI	0.022113
THOMPSON	1.590857
KG	0.005261
ENTROPY	4.794664
PORTFOLIO	1.003351

Iteration 21

Method	Max Value
UCB	0.982773
EI	0.003868
PI	0.017397
THOMPSON	1.844622
KG	0.003868
ENTROPY	4.355976
PORTFOLIO	1.094365

Iteration 22

Method	Max Value
UCB	0.978517
EI	0.003554
PI	0.016438
THOMPSON	1.362201
KG	0.003554
ENTROPY	3.796858
PORTFOLIO	1.010292

Iteration 23

Method	Max Value
UCB	0.967186
EI	0.003182
PI	0.015096
THOMPSON	1.676754
KG	0.003182
ENTROPY	3.951336
PORTFOLIO	1.114566

Iteration 24

Method	Max Value
UCB	0.934779
EI	0.002493
PI	0.012282
THOMPSON	1.319521
KG	0.002493
ENTROPY	4.398633
PORTFOLIO	1.038788

Iteration 25

Method	Max Value
UCB	0.928642
EI	0.002320
PI	0.011632

THOMPSON	1.895441
KG	0.002320
ENTROPY	4.219332
PORTFOLIO	0.955421

Iteration 26

Method	Max Value
UCB	0.917733
EI	0.002089
PI	0.010688
THOMPSON	1.391543
KG	0.002089
ENTROPY	3.912285
PORTFOLIO	0.965343

Iteration 27

Method	Max Value
UCB	0.916993
EI	0.002077
PI	0.010652
THOMPSON	1.459246
KG	0.002077
ENTROPY	3.730572
PORTFOLIO	1.024184

Iteration 28

Method	Max Value
UCB	0.910246
EI	0.001868
PI	0.009840
THOMPSON	1.256596
KG	0.001868
ENTROPY	3.674764
PORTFOLIO	0.935906

Iteration 29

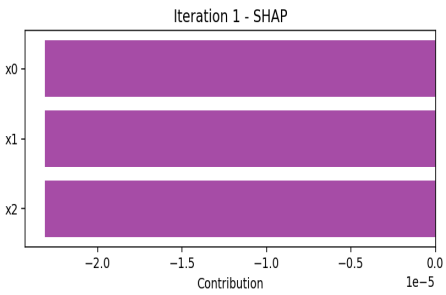
Method	Max Value
UCB	0.917303
EI	0.001912
PI	0.010123
THOMPSON	1.571633
KG	0.001912
ENTROPY	3.746246
PORTFOLIO	0.949660

Iteration 30

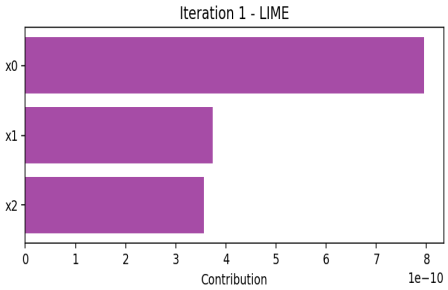
Method	Max Value
UCB	0.889939
EI	0.001562
PI	0.008470
THOMPSON	1.533703
KG	0.001562
ENTROPY	3.816998
PORTFOLIO	0.815827

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



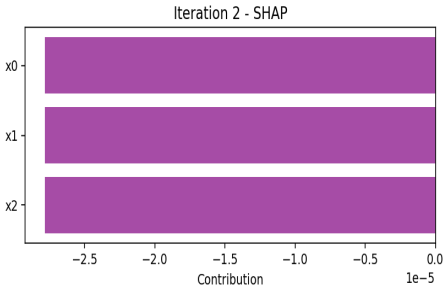
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	-0.00002
x1	-0.00002
x2	-0.00002

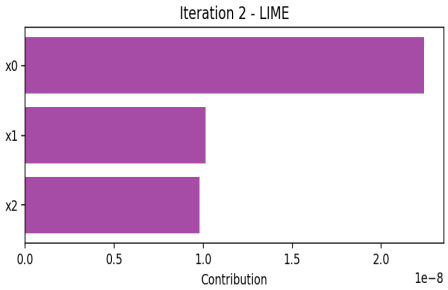
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 2 - SHAP



Iteration 2 - LIME



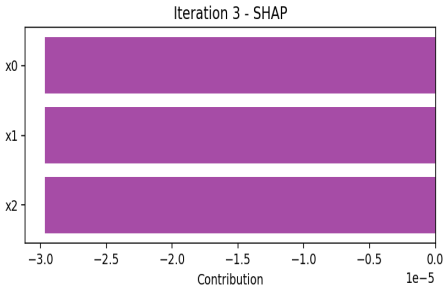
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	-0.00003
x1	-0.00003
x2	-0.00003

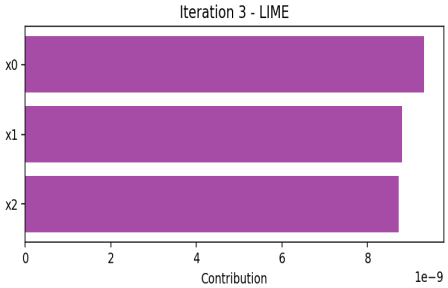
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 3 - SHAP



Iteration 3 - LIME



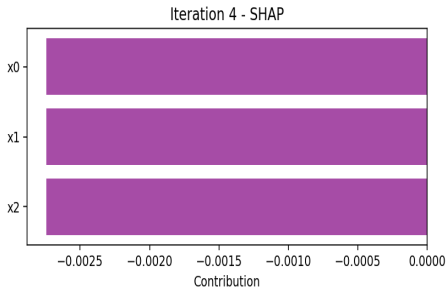
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-0.00003
x1	-0.00003
x2	-0.00003

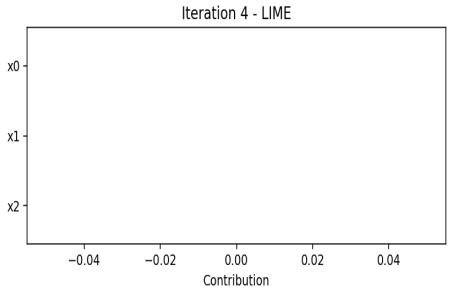
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 4 - SHAP



Iteration 4 - LIME



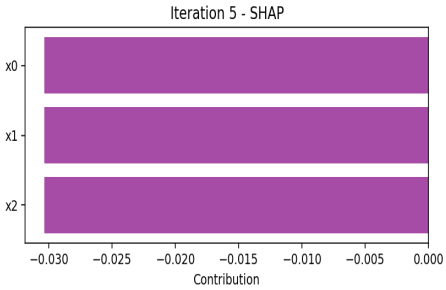
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.00274
x1	-0.00274
x2	-0.00274

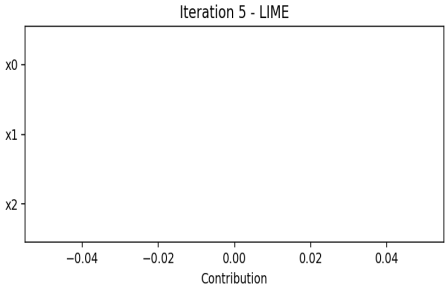
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 5 - SHAP



Iteration 5 - LIME



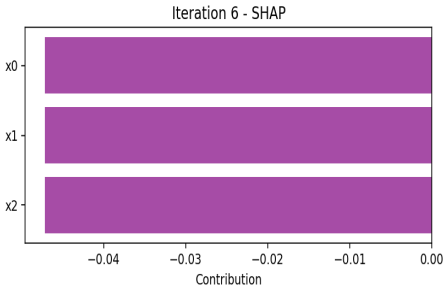
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.03035
x1	-0.03035
x2	-0.03035

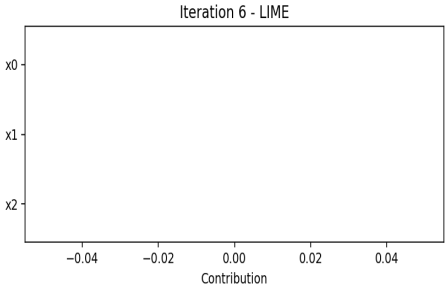
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 6 - SHAP



Iteration 6 - LIME



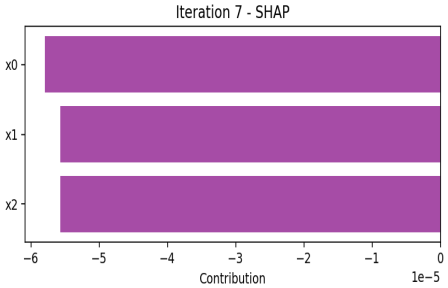
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.04717
x1	-0.04717
x2	-0.04717

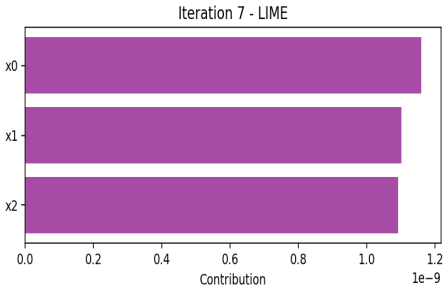
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 7 - SHAP



Iteration 7 - LIME



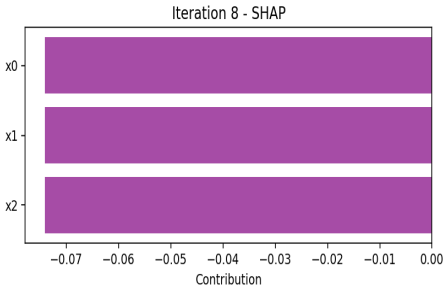
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.00006
x1	-0.00006
x2	-0.00006

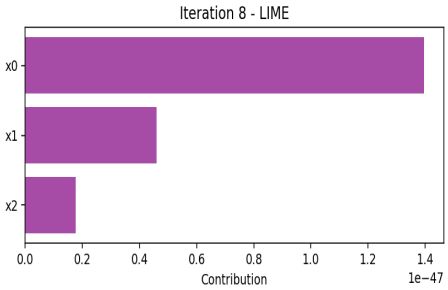
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 8 - SHAP



Iteration 8 - LIME



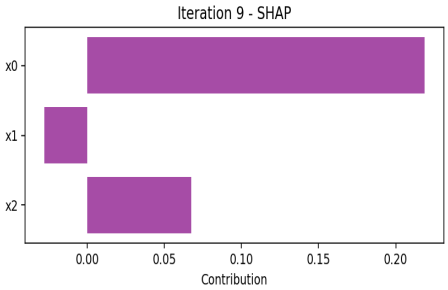
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	-0.07412
x1	-0.07412
x2	-0.07412

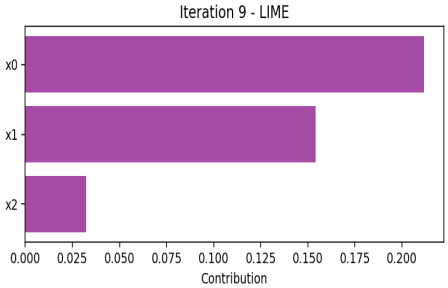
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.00000
x1	0.00000
x2	0.00000

Iteration 9 - SHAP



Iteration 9 - LIME



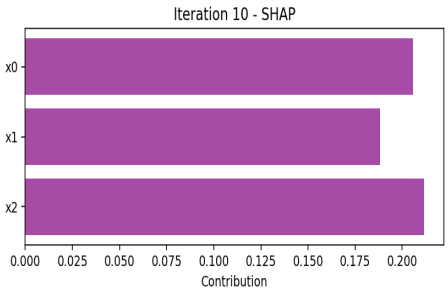
Iteration 9 - SHAP Values

Feature	SHAP Value
x0	0.21879
x1	-0.02775
x2	0.06751

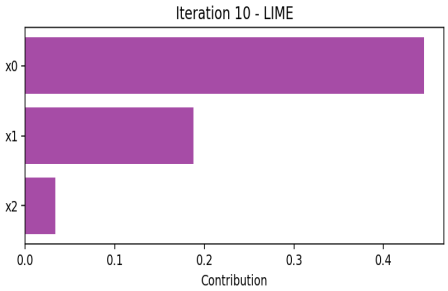
Iteration 9 - LIME Values

Feature	LIME Value
x0	0.21165
x1	0.15418
x2	0.03222

Iteration 10 - SHAP



Iteration 10 - LIME



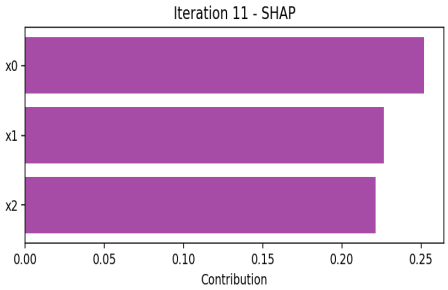
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.20558
x1	0.18814
x2	0.21146

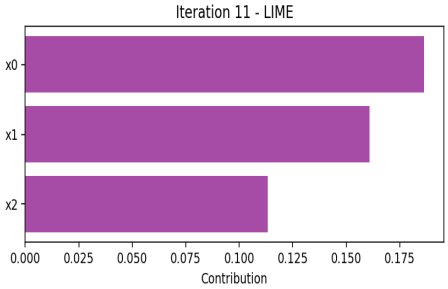
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.44498
x1	0.18790
x2	0.03332

Iteration 11 - SHAP



Iteration 11 - LIME



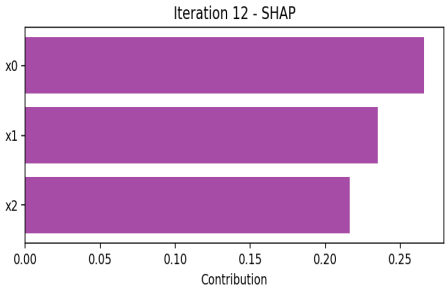
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	0.25176
x1	0.22660
x2	0.22126

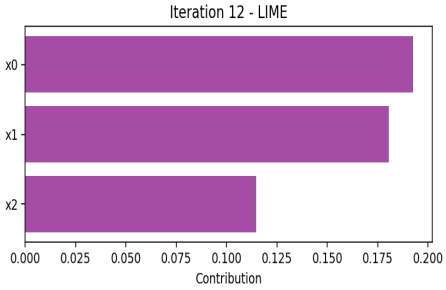
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.18632
x1	0.16104
x2	0.11346

Iteration 12 - SHAP



Iteration 12 - LIME



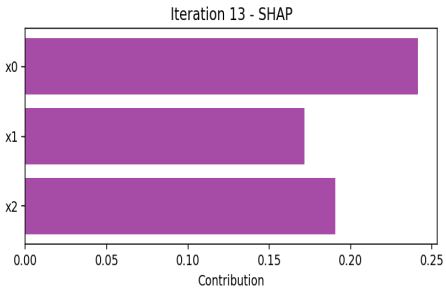
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	0.26569
x1	0.23498
x2	0.21612

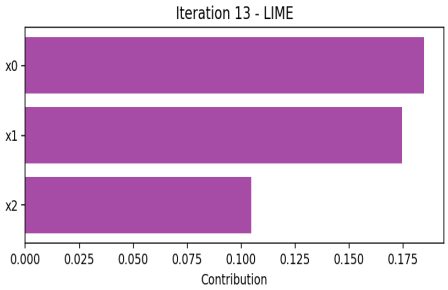
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.19255
x1	0.18051
x2	0.11466

Iteration 13 - SHAP



Iteration 13 - LIME



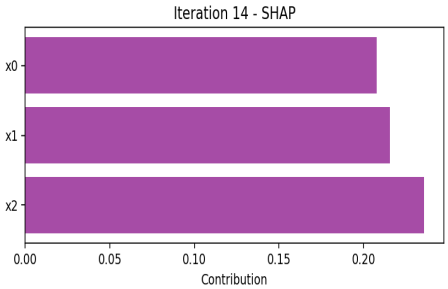
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.24140
x1	0.17150
x2	0.19050

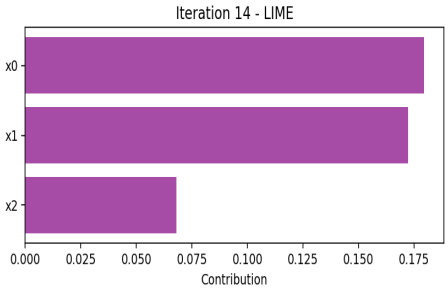
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.18462
x1	0.17459
x2	0.10471

Iteration 14 - SHAP



Iteration 14 - LIME



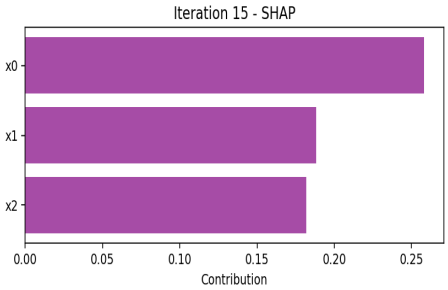
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.20782
x1	0.21553
x2	0.23574

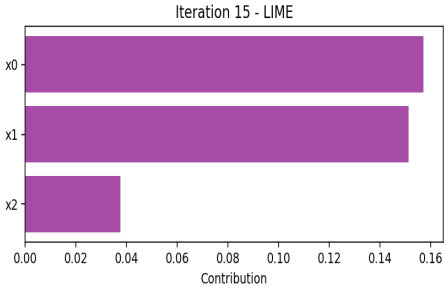
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.17939
x1	0.17239
x2	0.06796

Iteration 15 - SHAP



Iteration 15 - LIME



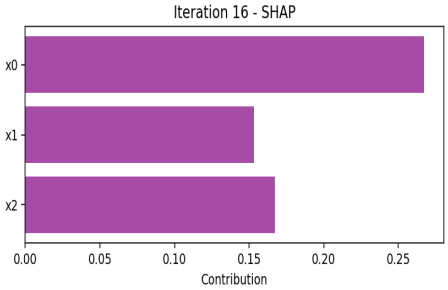
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	0.25800
x1	0.18844
x2	0.18184

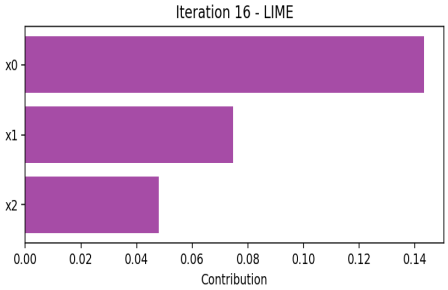
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.15715
x1	0.15133
x2	0.03745

Iteration 16 - SHAP



Iteration 16 - LIME



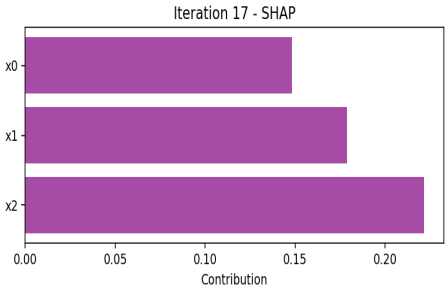
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	0.26710
x1	0.15310
x2	0.16734

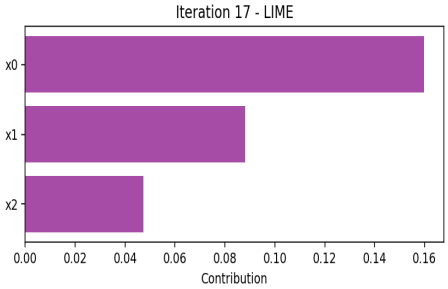
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.14329
x1	0.07478
x2	0.04795

Iteration 17 - SHAP



Iteration 17 - LIME



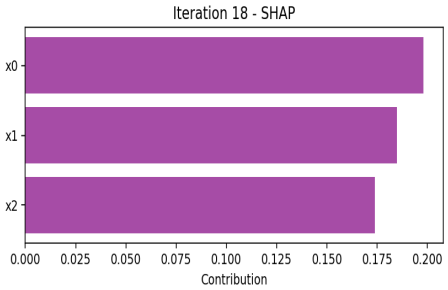
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	0.14824
x1	0.17873
x2	0.22153

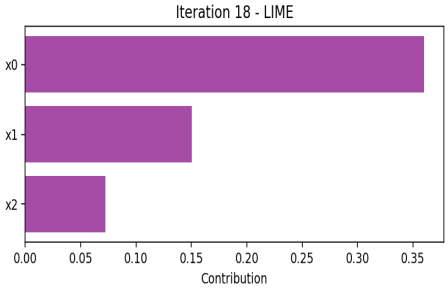
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.15974
x1	0.08816
x2	0.04729

Iteration 18 - SHAP



Iteration 18 - LIME



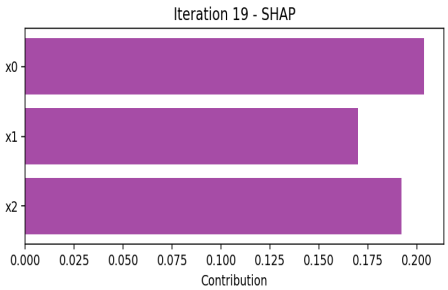
Iteration 18 - SHAP Values

Feature	SHAP Value
x0	0.19800
x1	0.18488
x2	0.17387

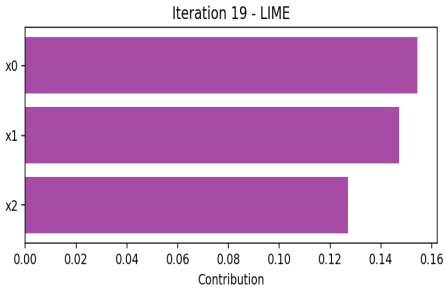
Iteration 18 - LIME Values

Feature	LIME Value
x0	0.35986
x1	0.15050
x2	0.07222

Iteration 19 - SHAP



Iteration 19 - LIME



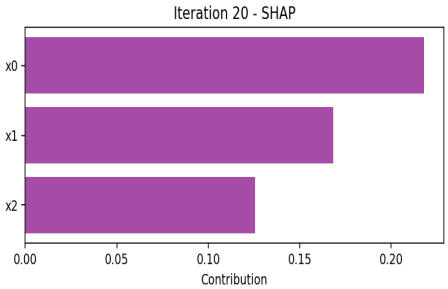
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	0.20367
x1	0.17002
x2	0.19220

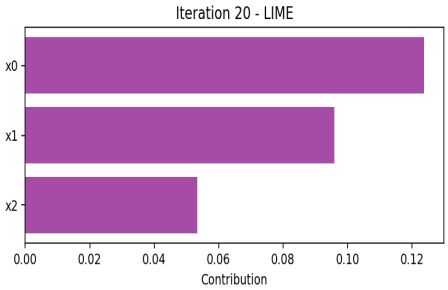
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.15443
x1	0.14709
x2	0.12701

Iteration 20 - SHAP



Iteration 20 - LIME



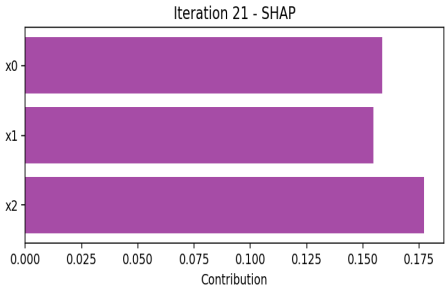
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	0.21783
x1	0.16833
x2	0.12568

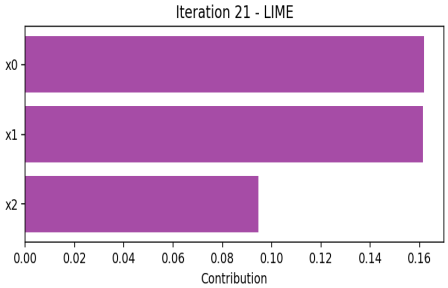
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.12372
x1	0.09593
x2	0.05331

Iteration 21 - SHAP



Iteration 21 - LIME



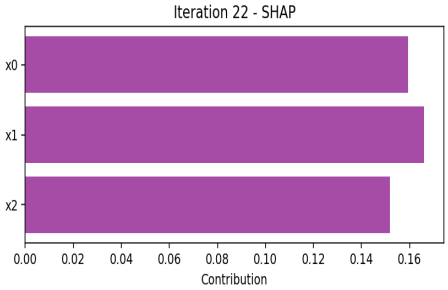
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	0.15859
x1	0.15462
x2	0.17713

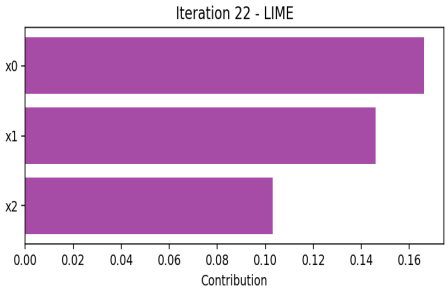
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.16182
x1	0.16136
x2	0.09454

Iteration 22 - SHAP



Iteration 22 - LIME



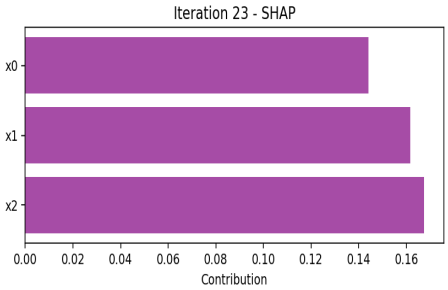
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	0.15945
x1	0.16599
x2	0.15182

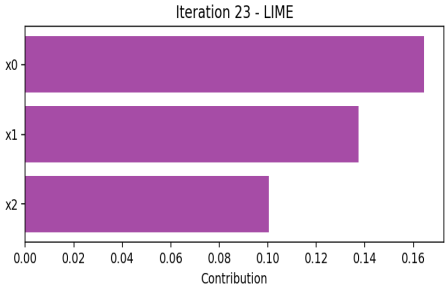
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.16602
x1	0.14600
x2	0.10309

Iteration 23 - SHAP



Iteration 23 - LIME



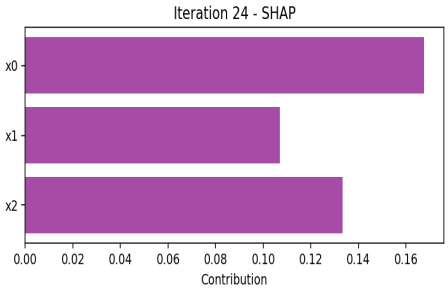
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	0.14408
x1	0.16169
x2	0.16735

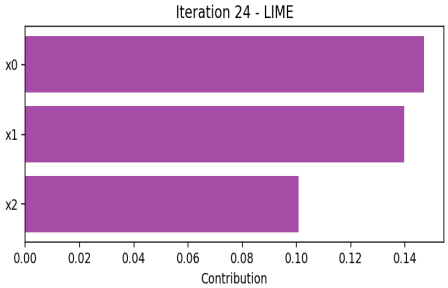
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.16432
x1	0.13732
x2	0.10046

Iteration 24 - SHAP



Iteration 24 - LIME



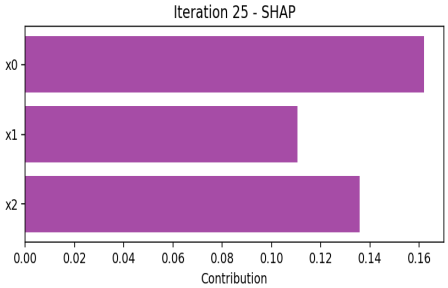
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	0.16750
x1	0.10686
x2	0.13336

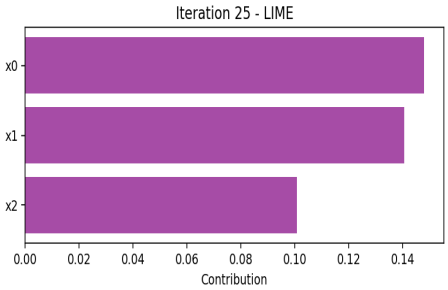
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.14696
x1	0.13970
x2	0.10083

Iteration 25 - SHAP



Iteration 25 - LIME



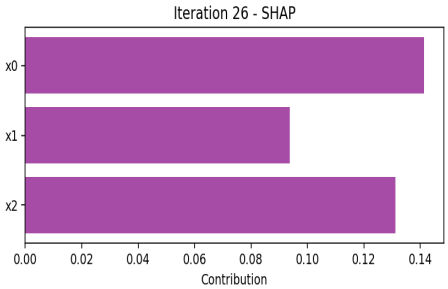
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	0.16187
x1	0.11048
x2	0.13569

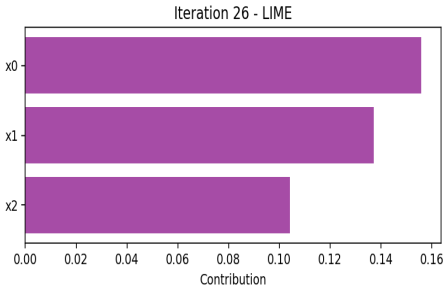
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.14792
x1	0.14072
x2	0.10088

Iteration 26 - SHAP



Iteration 26 - LIME



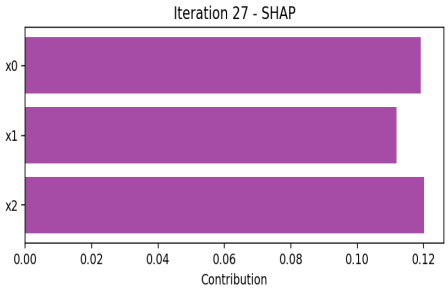
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	0.14123
x1	0.09370
x2	0.13116

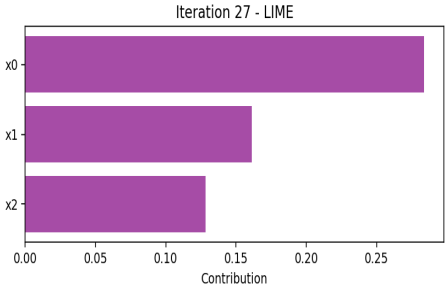
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.15588
x1	0.13724
x2	0.10414

Iteration 27 - SHAP



Iteration 27 - LIME



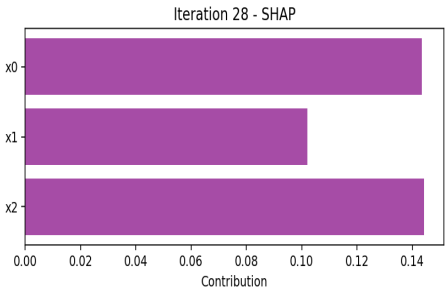
Iteration 27 - SHAP Values

Feature	SHAP Value
x0	0.11907
x1	0.11188
x2	0.12005

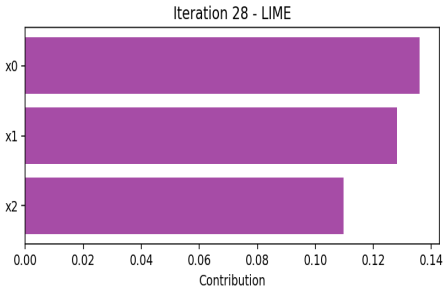
Iteration 27 - LIME Values

Feature	LIME Value
x0	0.28373
x1	0.16138
x2	0.12840

Iteration 28 - SHAP



Iteration 28 - LIME



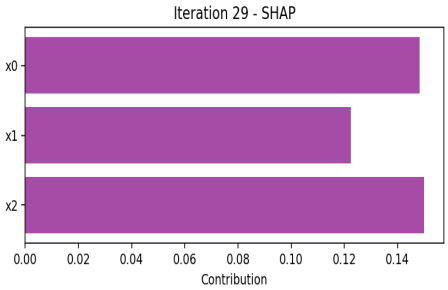
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	0.14351
x1	0.10211
x2	0.14422

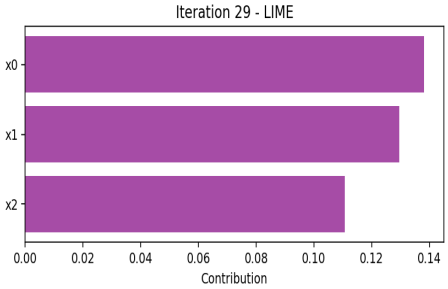
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.13602
x1	0.12822
x2	0.10969

Iteration 29 - SHAP



Iteration 29 - LIME



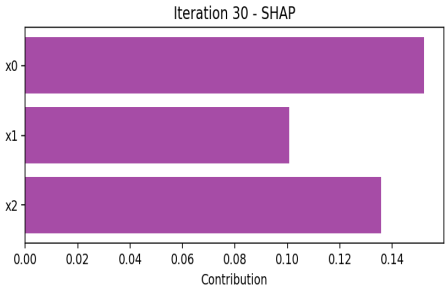
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	0.14837
x1	0.12240
x2	0.14992

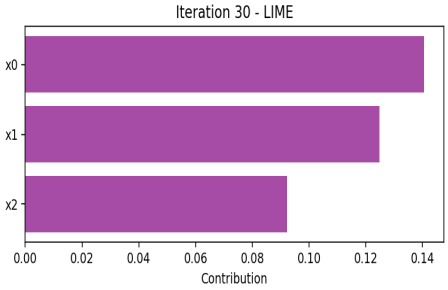
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.13812
x1	0.12955
x2	0.11068

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	0.15208
x1	0.10067
x2	0.13571

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.14044
x1	0.12484
x2	0.09226